

AFFAN KHAN

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PROFESSIONAL SUMMARY

Computer Science undergraduate (BSCS, FAST-NUCES) with hands-on experience in C++, object-oriented programming, and database design. Currently interning at Deister Software Pakistan, building HR database schemas in Informix using the XDBL framework. Built multiple independent projects spanning 2D game development and AI-powered web applications. Seeking opportunities to apply strong programming fundamentals to real-world software engineering challenges.

EDUCATION

Bachelor of Science in Computer Science (BSCS) | *FAST-NUCES, Islamabad* 2025 – 2029 (Expected)
Currently in 3rd Semester | CGPA: 3.6 / 4.0

PROFESSIONAL EXPERIENCE

Software Development Intern | *Deister Software Pakistan* 2026 – Present

- Designed and developed HR database schemas in Informix using Axional/Deister Studio, covering departments, employees, payroll, job postings, and applicants modules.
- Built schema definitions using the proprietary XDBL XML-based schema format, applying core relational database concepts including relationships, foreign keys, and CLOB/BLOB data types.
- Generated and validated structured seed data for HR modules through iterative testing and correction cycles.
- Collaborated within an enterprise database environment, gaining practical exposure to production-grade schema modeling and data integrity practices.

PROJECTS

Affinity — AI Chat Clone | *Personal Project*

- Built a web-based AI chat application that integrates a large language model (LLM) API to deliver conversational responses.
- Developed the front-end interface using HTML, CSS, and JavaScript, focusing on a clean, responsive chat experience.
- Implemented API request handling and dynamic response rendering to simulate real-time conversational AI interaction.

Tumblepop — 2D Arcade Game Recreation | *Personal Project* | C++, SFML

- Recreated the classic Tumblepop arcade game using C++ and the SFML graphics library, applying object-oriented programming principles.
- Designed class hierarchies for game entities (player, enemies, bubbles) using polymorphism and encapsulation for modular, maintainable code.
- Implemented collision detection, sprite animation, and level progression logic from scratch.

Metal Slug — 2D Side-Scrolling Action Game | *Personal Project* | C++, SFML

- Developed a 2D side-scrolling action game inspired by Metal Slug, built entirely in C++ with SFML.
- Applied OOP design patterns to structure player mechanics, enemy AI, weapons, and boss fight sequences.
- Debugged and optimized sprite hitboxes and collision systems for accurate, responsive gameplay.

SKILLS

Programming Languages: C++, C#, Python, SQL, HTML, CSS, JavaScript

Frameworks & Tools: React, SFML, Informix, XDBL, Axional/Deister Studio, Git, VS Code

Concepts: Object-Oriented Programming (OOP), Database Schema Design, 2D Game Engine Design, LLM/AI API Integration, Data Structures